Prism: Cross-Deal Fraud Detection Model

Mathematical specification, calibration and validation

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Saturday 2026-11-07 Decisions of record: `docs/MODEL_DESIGN.md`

Abstract

Prism holds coordinated fake loan applications before money is disbursed while leaving honest but unusual borrowers alone. Fraud rings manufacture applications cheaply by *reusing resources*: a bookkeeper, a remit account, a document generator, a deposit-shaping script. Each reuse leaves a trace that is *rare in the bank's own history* and *shared across deals*. Every piece of evidence is expressed in one unit, bits of surprise relative to a year of clean history. Evidence is grouped into independent families, and a hold requires agreement between at least three of them. Thresholds are set to fixed false-alarm rates on clean deals. They are never fitted to fraud examples, so the method carries over to an unseen trick. Every scored decision is deterministic; language models (NVIDIA Nemotron) work around the engine and never decide a hold.

1 Setup and notation

- $\mathcal{H}$: history deals ($N = |\mathcal{H}| = 257$, Nov 2025 – Oct 2026); $\mathcal{D}_t$: current deals whose documents have arrived by sim time t (files are visible only once $t \geq$ `arrive_at`).
- $\mathcal{C} \subset \mathcal{H}$: *clean calibration set*: funded, `fraud_label`=none, repayment current or paid off, applied on or before 2026-06-30.
- Entity types $\mathcal{T}$ = {preparer, preparer phone, registered agent, remit account, vendor, PDF producer, business address, bank, guarantor ID}. For deal d, $E(d)$ is its set of canonical entity values. Canonicalization casefolds, strips punctuation, reduces phones to digits, tokenizes addresses, and splits the legal suffix (LLC, Inc) into its own field, giving a *base key* and a *full key*.
- $\mathrm{df}(v) = |\{h \in \mathcal{H} : v \in E(h)\}|$, the history document frequency of entity value v.

2 The unit: bits of surprise

For any observation x with smoothed probability $\hat{P}(x \mid \text{clean history})$,

$$\mathcal{B}(x) \;=\; -\log_2 \hat{P}(x \mid \text{clean history}). \tag{1}$$

Every family $f \in \{1, \ldots, 6\}$ produces a per-deal score $\mathcal{B}_f(d) \geq 0$. All probabilities use add-one (Laplace) smoothing, so no observation is ever infinitely surprising. For an entity value,

$$\hat{p}(v) = \frac{\mathrm{df}(v) + 1}{N + 1}, \qquad I(v) = -\log_2 \hat{p}(v). \tag{2}$$

With $N = 257$: an unseen value carries $I \approx 8.0$ bits, a value used 5 times ≈ 5.4 bits, 15 times ≈ 4.0 bits, 200 times ≈ 0.4 bits.

3 Signal families

F1 Shared rare entity (mandatory for a hold)

An entity value v is *strong* if it is rare and actually links deals:

$$\text{strong}(v) \iff \frac{\text{df}(v)}{N} < \rho \;\wedge\; |\{d' \in \mathcal{H} \cup \mathcal{D}_t : v \in E(d')\}| \geq 2, \qquad \rho = 0.02. \tag{3}$$

The pair evidence between deals a and b weights each shared strong entity by its surprisal, its type and its history outcomes:

$$S(a,b) = \sum_{v \in E(a) \cap E(b),\ \text{strong}(v)} w_{t(v)}\, \omega_v\, I(v), \qquad \omega_v = 1 - \beta\, \frac{\text{repaid}(v)+1}{\text{df}(v)+2}. \tag{4}$$

Here w_t is the entity-type weight (remit account, preparer and registered agent strong; address and bank weak), and ω_v discounts entities whose history deals mostly repaid ($\beta \in [0,1]$). This is the rarity-weighted form of the standard *Adamic–Adar* link-prediction index $\text{AA}(a,b) = \sum_v 1/\log \deg(v)$, with $\deg(v) \approx \text{df}(v)$. The deal score is $\mathcal{B}_1(d) = \max_{b \neq d} S(d,b)$, over both current and history deals.

F2 Novelty and burst

For entity v, let λ_v be its smoothed long-run usage rate scaled to a recent window W (90 days), and k_v its uses inside the window. Under a Poisson null $X \sim \text{Pois}(\lambda_v)$,

$$\mathcal{B}_2(d) = \max_{v \in E(d)} -\log_2 P(X \geq k_v \mid \lambda_v), \qquad P(X \geq k) = 1 - \sum_{j=0}^{k-1} \frac{e^{-\lambda}\lambda^j}{j!}. \tag{5}$$

F3 Lookalike counterparty

Vendor u on deal d is a lookalike of a frequent history vendor u^* if their base keys match ($\text{sim}_{\text{token-set}} \geq 0.8$ or edit distance ≤ 2) but the suffix or the remit account differs. With n_{pat} the number of history equipment deals showing that same pattern, out of n_{vend} history vendor observations,

$$\mathcal{B}_3(d) = -\log_2 \frac{n_{\text{pat}}+1}{n_{\text{vend}}+1} \quad \text{if a lookalike exists, else 0.} \tag{6}$$

F4 Document forensics

For document class c (bank statement, tax return, ...) and producer string π, with $n_{c,\pi}$ observed counts and Π_c the producers seen for class c:

$$\hat{P}(\pi \mid c) = \frac{n_{c,\pi}+1}{n_c + |\Pi_c| + 1}, \qquad \mathcal{B}_4(d) = \max_{\text{doc} \in d} -\log_2 \hat{P}(\pi_{\text{doc}} \mid c_{\text{doc}}). \tag{7}$$

This flags a bank statement produced by a template tool instead of a statement system, without naming any tool.

F5 Deposit pattern

From statements, n deposits, of which R are multiples of \$100. A Beta(a, b) prior is fitted to the history round-share by the method of moments, and the shrunken share is

$$\tilde{r} = \frac{R + a}{n + a + b}, \tag{8}$$

which pulls small samples (e.g. four large wires) toward normal instead of using a hard cutoff. Weekday mix q (Dirichlet-smoothed) is compared with the pooled history weekday distribution p_0 through

$$H(q) = -\sum_i q_i \log_2 q_i, \qquad D_{\mathrm{KL}}(q \,\|\, p_0) = \sum_i q_i \log_2 \frac{q_i}{p_0(i)}. \tag{9}$$

Deposits only on two weekdays give $H = 1$ bit, while a spread over five gives about 2.3. Each statistic is turned into bits through its smoothed empirical upper tail on $\mathcal{C}$: $\mathcal{B}(z) = -\log_2 \frac{1+|\{h \in \mathcal{C} : z_h \geq z\}|}{1+|\mathcal{C}|}$, and $\mathcal{B}_5(d)$ is the maximum over the statistics.

F6 Terms shape (evidence only, never votes)

Amount within 2% below any approval threshold in `stages.json`, entity younger than 24 months, no needed-by date, round amount. Reported in the reason and the checklist, but excluded from the hold rule, because these traits are common among honest small businesses and correlate with other quirks.

4 Calibration: thresholds from clean deals only

For $f \in \{2, 3, 4, 5\}$ the firing threshold is the interpolated 95th percentile on the clean calibration set:

$$\tau_f = Q_{0.95}(\{\mathcal{B}_f(h) : h \in \mathcal{C}\}), \qquad \mathrm{fires}_f(d) \iff \mathcal{B}_f(d) \geq \tau_f, \tag{10}$$

so each family misfires on about $\alpha = 5\%$ of good deals by construction ($|\mathcal{C}| \approx 164$ makes a 5% tail about 8 deals, a stable estimate; a 1% tail would be 1–2 deals). F1 fires when $\mathcal{B}_1(d) \geq \tau_1$ with at least one strong shared entity. The configuration stores the *percentile* (0.95) and *fraction* (ρ), never bit values, so thresholds re-derive from whatever history ships with a new dataset. The fraud examples and the 13 sample deals are *never* used to set a threshold.

5 Decision rule

$$\textsc{Hold}(d, t) \iff \mathrm{fires}_1(d, t) \wedge \sum_{f=2}^{5} \mathrm{fires}_f(d, t) \geq 2, \tag{11}$$

$$\textsc{Watch}(d, t) \iff 1 \leq \sum_{f=1}^{5} \mathrm{fires}_f(d, t) \wedge \neg\textsc{Hold}(d, t). \tag{12}$$

A hit on any of the bank's five known-fraud checks goes straight to Fraud Operations (bank policy).

False-alarm bound. If families misfire independently with rates α_1 and $\alpha = 0.05$,

$$P(\textsc{Hold} \mid \text{clean}) \le \alpha_1 \cdot P\big(\mathrm{Bin}(4, 0.05) \ge 2\big) = \alpha_1\big(1 - 0.95^4 - 4(0.05)(0.95)^3\big) \approx 0.014\,\alpha_1. \quad (13)$$

With $\alpha_1 \approx 0.05$ this is about 7×10^{-4} per deal: about 0.9% chance of any false hold over 13 deals, and an expected 0.07 false flags over the 93 held-out history deals. Families are not truly independent (that is why F6 cannot vote), so this bound is a design target. The real rate is *measured* on held-out history.

Display score. The UI shows total evidence $\mathcal{B}(d) = \sum_f \mathcal{B}_f(d)$ bits, read as "about 1 in $2^{\mathcal{B}}$ by chance under the clean-history model". The hold decision is always the rule above, never this number.

6 Clusters, significance and linking

- **Clusters**: connected components of the graph restricted to strong F1 edges. Dropping weak edges keeps common entities from fusing everything.
- **Permutation test**: for cluster K with statistic $T(K) = \sum_{a<b\in K} S(a, b)$, permute entity values within each type across all deals (preserving frequencies) $M = 1000$ times. Then $p = \frac{1+|\{m:T^*_m \ge T(K)\}|}{1+M}$.
- **Kit detection (PMI)**: for rare entities v, w, $\mathrm{PMI}(v, w) = \log_2 \frac{\hat{P}(v,w)}{\hat{P}(v)\hat{P}(w)}$. High-PMI pairs mark a reused kit (preparer + account + producer together).
- **History flags**: history deal h joins `history_flags` iff it has at least 2 independent links to a held cluster, from: (a) a strong shared entity type, (b) a second strong type, (c) the same F3 pattern, (d) the same rare producer, (e) a deposit fingerprint within a KL bound.

7 Timing

Scores are recomputed on every file arrival. With $\mathcal{A}$ the arrival times,

$$\texttt{flagged_at}(d) = \min\{t \in \mathcal{A} : \textsc{Hold}(d, t)\}. \quad (14)$$

Holds are *sticky*: once requested, a deal stays held until a human sends `fraud_release`. Weaker later evidence adds a log entry and never un-flags.

8 Validation design

Set	Size	Role	Tuned on?
History Nov 2025 – Jun 2026	164	baselines, thresholds	yes (only set)
History Jul – Oct 2026, forward replay	93	held-out false flags, scheme position	no
Old-trick fraud deals	12	blind check without rules	no
Sample deals	13	regression, trap suite	no
Synthetic tricks + benign twins	100s	leave-one-family-out	sanity only
Nemotron-invented tricks	—	final exam	never
Weekend drop	?	blind, first run recorded	never

Success, in priority order: (1) zero false holds on the sample traps and zero history false flags; (2) the sample ring held before Closed, with exact `history_flags`; (3) held-out history: at most 1 flag among the good deals, each reviewed in writing; (4) synthetic held-out family: at least 70% of ring deals held, at most 5% of benign twins flagged; (5) earliness. A lower criterion never overrides a higher one.

9 Playbooks

- **Ring missed**: fix extraction → generic feature fix (full re-run) → threshold change only if held-out still passes (logged). Time box 60 minutes; otherwise ship it on watch and say so. Never sample literals or per-deal exceptions.
- **False hold**: inspect the families → lower the type weight / add the outcome discount / fix the feature → tighten to F1 + 3 of F2–F5. Never whitelist by name.
- **Running late**: build F1, F3, F5, F4, F2, then PMI/permutation. F1 + F3 + F5 alone can satisfy the rule.

10 Nemotron guardrails

A Nemotron vision read $\hat{x}$ may fill a scored field x only if

$$x_{\text{parser}} = \varnothing \ \wedge \ \text{fmt}(\hat{x}) \ \wedge \ \left(\nexists\, x_{\text{2nd}} \ \vee \ \hat{x} \equiv x_{\text{2nd}}\right), \tag{15}$$

and it can never create an issue, mismatch or hold. Calls are cached by hash(model, messages, tools, parameters), so runs replay deterministically. Agents (adaptive red-team, prosecutor/defender) use read-only HTTP tools; every claim must cite an evidence ID, and uncited claims are flagged by a deterministic check.

A UI handoff (for Jesse's agents)

The full agent-ready playbook is `docs/UI_PLAYBOOK.md`; the locked visual decisions are in `.planning/phases/02-*/02-CONTEXT.md` (D-01 to D-16). To set up Claude Code on the Mac, inside the repo:

```
# inside Claude Code (frontend-design + webapp-testing skills):
/plugin marketplace add anthropics/skills
/plugin install example-skills@anthropic-agent-skills
# in a terminal (shadcn component MCP; Playwright MCP so the agent sees the page):
npx shadcn@latest mcp init --client claude
claude mcp add playwright -- npx @playwright/mcp@latest
```

How this model shows up on screen: the evidence panel lists the families in order F1–F6 with their bits and thresholds; F6 is visually marked "evidence only". Edges in the graph are weighted by $I(v)$ (thicker means rarer), and common entities are drawn faint with their history count ("used by 15 deals, mostly repaid"). The cluster header shows the permutation p-value and "k of 4 voting families agree".